

AI-assisted Portfolio Project Report

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1. Project Overview

Objective: The objective of this project was to create a professional online CV and portfolio with interactive features and GitHub API integration, using AI coding tools to assist in development.

Features Implemented:

- HTML CV structure and semantic sections (About, Experience, Projects, Skills, Contact).
- Responsive CSS styling with light/dark themes and accessible components.
- JavaScript interactivity: navigation highlighting, section toggles, project detail expansion, dynamic skills addition, contact form validation, and a print-based "Download CV" action.
- GitHub API integration to fetch and display public repositories dynamically.
- Documentation and deployment instructions (README, AI log).

2. AI Tool(s) Used

Tool Name	Purpose / Task Assisted
GitHub Copilot (VS Code)	Generated HTML snippets, CSS patterns, and JavaScript helper functions used throughout the site.
OpenAI Copilot Chat / ChatGPT	Reviewed code, suggested refactors, and produced the LaTeX report and additional documentation content.

3. Tasks Assisted by AI

1. **HTML/CSS scaffolding:** AI provided initial layout and style suggestions; I edited markup for semantics and accessibility.
2. **JavaScript functions:** AI suggested patterns for event handling and API integration; I improved validations and error handling.
3. **LaTeX report generation:** AI produced the report template and filled content which I adapted and verified.

4. Challenges Encountered

AI-generated suggestions were occasionally incomplete (edge-case handling) or stylistically inconsistent with existing code. Debugging and testing were necessary to ensure the features worked reliably across browsers.

5. Reflections on AI Usage

1. What worked well: Rapid scaffolding, repetitive code generation, and idea exploration.

2. What needed manual correction: Accessibility attributes, error handling, and API resilience.
3. Productivity: AI accelerated initial development, allowing focus on integration and polish.
4. Understanding: Using AI required careful review which reinforced my knowledge of event-driven JS and responsive CSS.
5. Future use: I would continue using AI tools as assistants but maintain manual verification and testing.

6. Ethical and Practical Considerations

Relying on AI raises concerns about correctness, licensing of generated snippets, and potential over-dependence. Developers must verify generated code, attribute third-party content properly, and avoid using AI as a substitute for critical thinking.

7. Online Portfolio Link

Hosted URL

8. Conclusion

This AI-assisted project provided practical experience using code-completion and chat-based AI tools to implement a professional portfolio. The tools improved speed but required hands-on validation for correctness and accessibility. Future improvements include stronger API error handling, adding tests, and deploying the site publicly.